
EV Charging Station Analytics

An End-to-End Data Analytics Project using SQL, Excel & Power BI

Abstract

This project analyzes the operations of an electric vehicle (EV) charging network spanning 10 cities, 50 charging stations, 3,000 customers, 3,000 vehicles, and over 30,000 charging sessions. The dataset captures station-level attributes, customer demographics, vehicle specifications, and detailed session-level metrics such as duration, energy consumed, tariff, revenue, payment method, weather, and traffic conditions. Using SQL for data modeling and querying, Excel for exploratory analysis, and Power BI for interactive visualization, the project derives actionable insights into revenue trends, station performance, customer charging behavior, and the influence of external factors such as weather and traffic on charging demand.

Introduction

The rapid adoption of electric vehicles has made charging infrastructure a critical area of study for operators, city planners, and investors. Understanding how stations are used, which factors drive revenue, and how customer and vehicle characteristics relate to charging patterns helps stakeholders optimize station placement, pricing, and resource allocation. This project simulates a realistic EV charging ecosystem across a relational database of six interconnected entities: cities, stations, customers, vehicles, and charging sessions. The goal is to build a complete analytics workflow — from raw relational data to a polished, interactive Power BI dashboard — that mirrors a real-world data analyst engagement, and to publish the full workflow as an open, reproducible project on GitHub.

Tools Used

Tool	Purpose in this Project
MySQL / SQL	Database design (schema, primary/foreign keys), data storage, and exploratory queries: KPIs, top-N rankings, group-by aggregations, and trends.
Microsoft Excel	Initial data inspection, cleaning checks, pivot tables, and quick cross-verification of SQL query results.
Power BI	Data modeling (relationships between fact and dimension tables), DAX measures, and an interactive multi-page dashboard with slicers and visuals.
CSV Data Files	Source data for cities, stations, customers, vehicles, and charging sessions, used to populate the SQL database and Power BI model.

Steps Involved in Building the Project

- **1. Data Preparation:** Collected and structured raw data into five entities — cities, stations, customers, vehicles, and charging sessions — stored as CSV files.
- **2. Database Design (SQL):** Designed a relational schema in MySQL with primary keys for each table and foreign keys linking stations to cities, vehicles to customers, and charging sessions to customers, vehicles, and stations.
- **3. Data Loading & Validation:** Loaded the CSV data into the SQL tables and validated referential integrity and row counts across all five tables.
- **4. Exploratory SQL Analysis:** Wrote analytical queries to compute total sessions, total revenue, total energy consumed, average charging duration, top revenue-generating stations, sessions by charger type, revenue by payment method, sessions by weather, top customers by energy usage, average revenue by charger type, monthly revenue trends, EV brand-wise sessions, average energy by vehicle type, and top cities by session count.

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- **5. Excel-Based Cross-Checks:** Used Excel pivot tables and formulas to independently verify key SQL results and to perform quick ad-hoc exploration of the raw data.
 - **6. Data Modeling in Power BI:** Imported all five tables into Power BI and built a star-schema style data model, establishing relationships between the charging sessions fact table and the cities, stations, customers, and vehicles dimension tables.
 - **7. Dashboard Development:** Created DAX measures for KPIs (total sessions, total revenue, total energy, average duration) and built interactive visuals: revenue trends over time, top stations and cities, charger-type and payment-method breakdowns, and the impact of weather and traffic on charging activity.
 - **8. Insight Generation:** Interpreted the visuals and query outputs to identify high-performing stations, popular charger types, peak revenue periods, and behavioral patterns among customers and EV brands.
 - **9. Documentation & Publishing:** Compiled the SQL scripts, datasets, Power BI file, and this report into a single project folder for publishing on GitHub as a portfolio-ready, reproducible data analytics project.

Conclusion

This project demonstrates a complete data analytics workflow — from relational database design through SQL analysis to an interactive Power BI dashboard — applied to a realistic EV charging network. The combined use of SQL, Excel, and Power BI enabled efficient data validation, deep exploratory analysis, and clear, decision-ready visualizations covering revenue performance, station utilization, customer behavior, and external factors such as weather and traffic. Beyond the specific findings, the project showcases practical, industry-relevant skills in database design, query writing, and dashboard building, and is structured for easy sharing and review as an open-source portfolio project on GitHub.